TOMMI PICOULET-GIUSTA

Software Developer — C/C++, Systems & Full-Stack Web

Nice, France | +33 6 14 63 09 14 | tpicoule123@gmail.com | github.com/Tpicoulegiusta | linkedin.com/in/tommi-picoulet-2571a7405

SUMMARY

Software developer trained at École 42 Nice, a selective project-based coding school with no formal classes, where students build production-grade software peer-to-peer. Strong foundation in low-level C/C++, network programming and concurrency, plus full-stack web with React and Node.js. Expected graduation December 2028. Open to relocation anywhere in Canada, seeking a software development co-op internship of 4 to 12 months, eligible under the International Experience Canada (International Co-op) work permit.

TECHNICAL SKILLS

Languages: C, C++, x86-64 Assembly (NASM), Python, JavaScript, TypeScript, Bash

Web: React, Node.js, HTML, CSS, REST APIs, WebSockets

Systems & Tools: Linux, Git/GitHub, Docker, Docker Compose, Make, GDB, Valgrind

Concepts: Multithreading & synchronization, socket programming, manual memory management, client-server architecture, object-oriented design

SOFTWARE PROJECTS

ft_transcendence — Real-Time Multiplayer Web Platform — React, Node.js, Docker

- Built a real-time multiplayer Pong platform with live chat, user accounts and matchmaking, delivered by a team of 5 developers using Git branching and code review.
- Implemented WebSocket game synchronization and containerized the full stack with Docker Compose for reproducible deployment.

IRC Server — RFC 1459 Compliant — C++

- Implemented an IRC server from scratch handling multiple concurrent clients over non-blocking sockets with a single-threaded poll loop, with no external libraries.
- Built full channel management, operator privileges, authentication and the standard command set, validated against real IRC clients.

Minishell — Unix Shell — C

- Recreated a working Unix shell: lexer/parser, pipes, input-output redirections, environment variables, signal handling and built-in commands.
- Managed process creation and memory manually with zero leaks, verified with Valgrind.

Cub3D — 3D Rendering Engine — C, MiniLibX

- Developed a raycasting 3D engine rendering a textured environment in real time, deriving the camera model and ray-geometry maths, with collision detection.

libasm — Standard Library in Assembly — x86-64 Assembly (NASM)

- Implemented core libc functions (strlen, strcpy, strcmp, write, read, strdup) in hand-written x86-64 assembly, following the System V AMD64 calling convention and handling errno on syscall failure.

NetPractice — Network Configuration — TCP/IP

- Diagnosed and configured ten progressively harder broken network topologies: IP addressing, subnet masks, default gateways, routing and OSI layers.

Inception — Containerized Infrastructure — Docker, NGINX, MariaDB

- Built a complete multi-service infrastructure with Docker Compose from custom images — NGINX with SSL/TLS termination, WordPress with php-fpm and MariaDB — configuring container networking, persistent volumes and secure deployment.

Philosophers — Concurrency Simulation — C

- Solved the dining philosophers problem with threads and mutexes, achieving deadlock-free shared-resource access under strict real-time timing constraints.

PROJECTS IN PROGRESS

Snow_crash — Security CTF — Reverse Engineering

- Working through a wargame of binary exploitation and privilege-escalation challenges: reverse engineering, system vulnerability analysis.

Music_Room — Shared Listening App — Web / Mobile

- Building a web and mobile application for real-time shared listening rooms with synchronized playback.

EDUCATION

École 42 Nice — Software Engineering — 2022 – December 2028 (expected)

- Peer-to-peer, project-based curriculum; no lectures, no teachers, evaluation by peer code review. Completed the Common Core at 100%; currently in the specialization track. Six projects validated with distinction ("Pass with bonus"), including Cub3D, the IRC server and ft_transcendence.
- Working toward RNCP Level 6 (French equivalent of a bachelor's degree), then Level 7 (master's equivalent). Expected graduation: December 2028.

Kaplan English School — Upper-Intermediate English — 2023

Baccalauréat, Lycée Masséna, Nice — 2017

WORK EXPERIENCE

Bartender — Hôtel Delcloy (3-star), Saint-Jean-Cap-Ferrat — June 2018 – November 2021

- Delivered high-volume service in peak season; teamwork, customer relations and composure under pressure.

Municipal Security Officer (ASVP) — Èze Village — Summer 2018; Summer 2026

- Sworn public officer: enforced municipal procedures with strict accuracy and daily contact with the public.

LANGUAGES

French — Native | English — C1 (Advanced) | Spanish — B2 (Intermediate)